

Transportation: Powered by EV

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DATA ANALYTIC ABOUT LAND TRANSPORTATION IN ASEAN
NATIONS AND THE UPCOMING EV(ELECTRIC VEHICLES)
MARKET IN THE REGION.

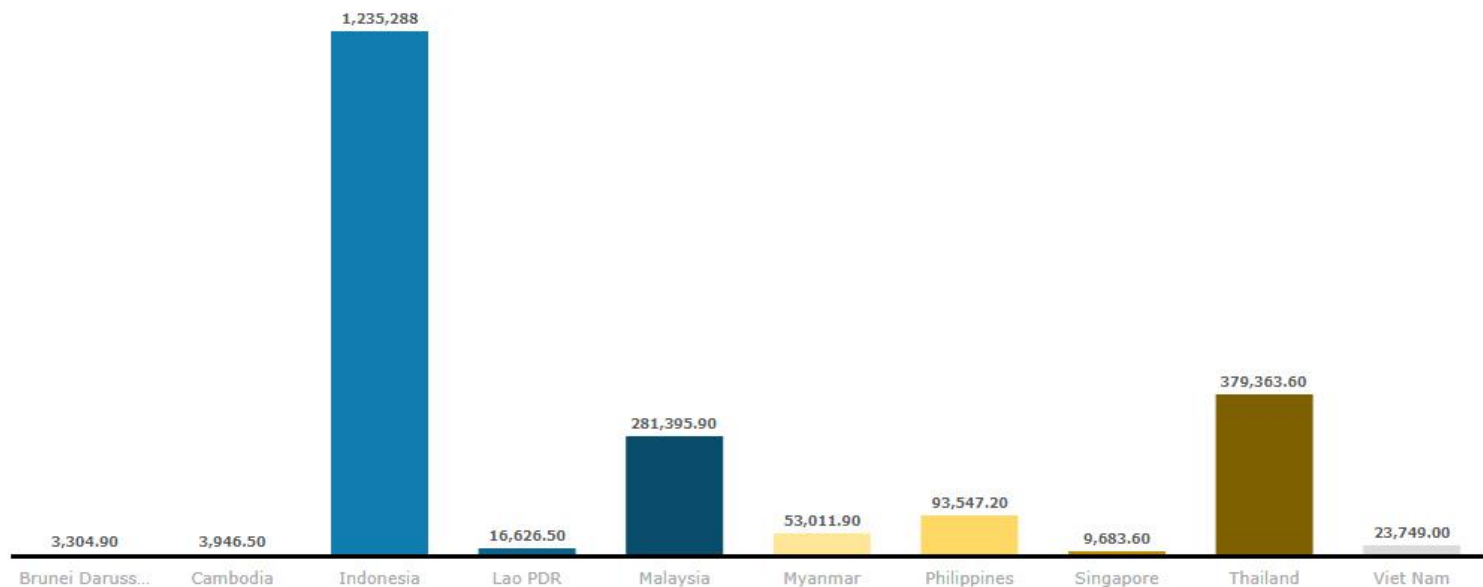
11 SUSTAINABLE CITIES
AND COMMUNITIES



Total Number of Registered Vehicle on Road [2012-2021]
(Thousand)

Source: ASEAN STATISTICAL YEARBOOK 2022

Brunei Darussalam Cambodia Indonesia Lao PDR Malaysia Myanmar Philippines Singapore Thailand Viet Nam

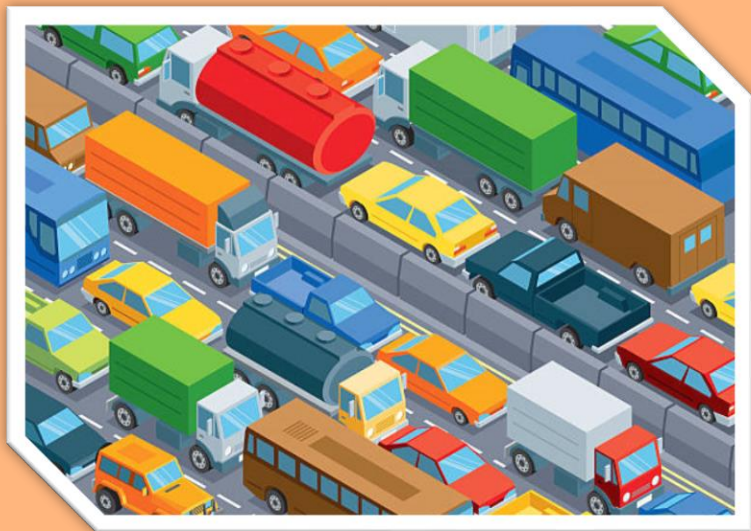


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Motorcycles are the most common vehicle in ASEAN.



Source : ASEAN Fuel Economy Roadmap for the Transport Sector 2018-2025: with Focus on Light-Duty Vehicles



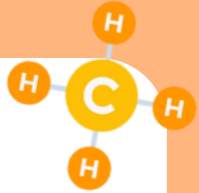
“In 2015, the ASEAN market was the **sixth-largest** vehicle market in the world.”

Source: ASEAN Fuel Economy Roadmap for the Transport Sector 2018-2025: with Focus on Light-Duty Vehicles

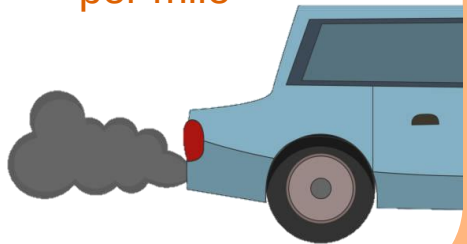




Greenhouse gases emission from vehicle: **carbon dioxide** (CO₂), **methane** (CH₄), **nitrous oxide** (N₂O) and **HCF**.

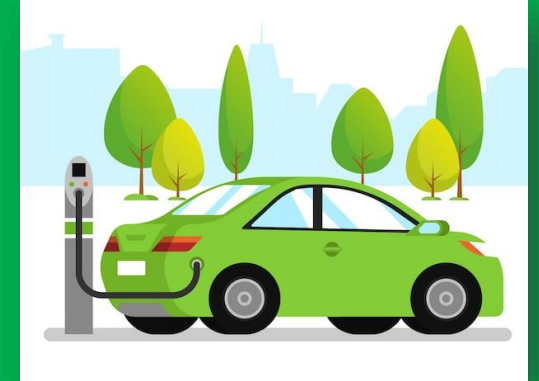


Average passenger vehicle emits **404** grams of CO₂ per mile



Source: Greenhouse Gas Emissions from a Typical Passenger Vehicle

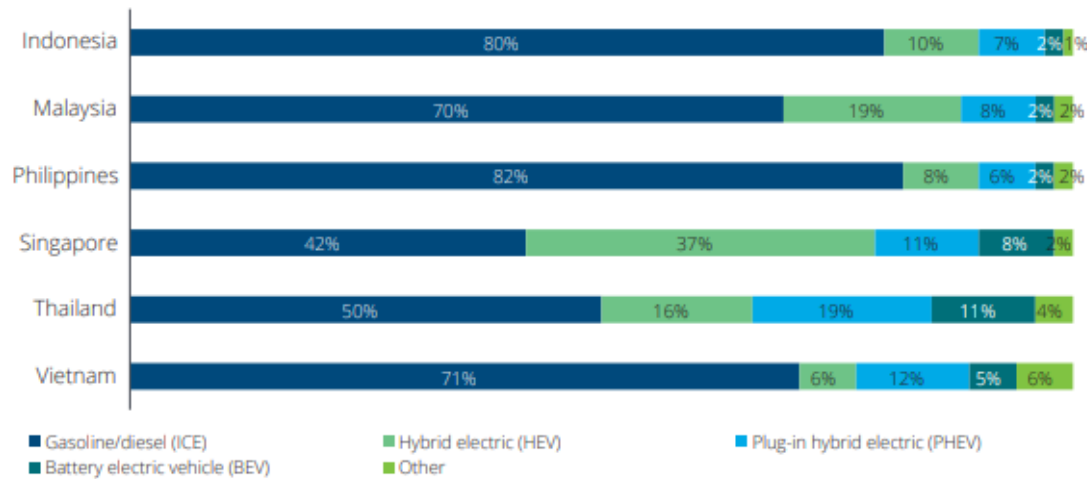
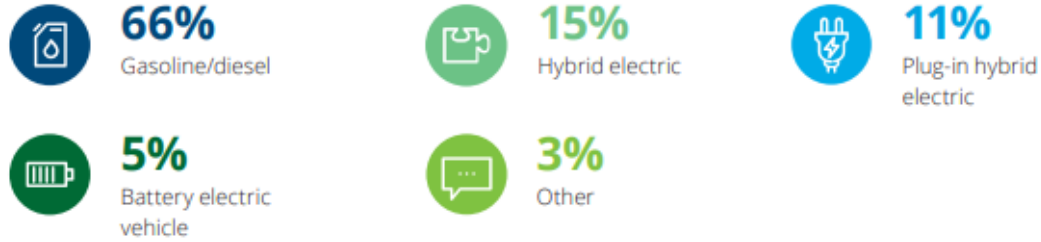
On average, driving an EV produces **50-60% fewer** greenhouse gas emissions than driving a gasoline-powered vehicle.



Source: EV for Anti-Pollution Matter



Overall powertrain preferences of Southeast Asian consumers



Note: "Other" includes engine types such as compressed natural gas, ethanol, and hydrogen fuel cells; did not consider "don't know" responses
 Q25. What type of engine would you prefer in your next vehicle?
 Sample size: n=5,070

Source: 2022 Global Automotive Consumer Study Southeast Asia Perspectives



Average price
of EV in ASEAN
USD 50,000 ~
100,000

Source: Electrical Vehicle Adoption
in ASEAN; Prospect and Challenges

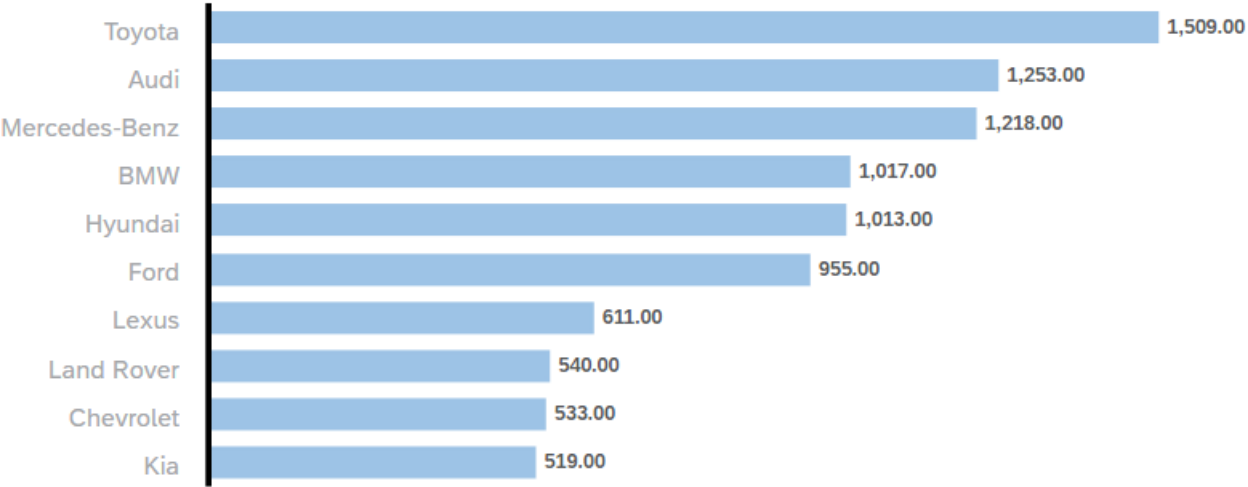
Source: 2022 Global Automotive Consumer Study Southeast Asia Perspectives

“The preference for specific types of EVs also varies across the region: interest in **BEVs** and **PHEVs** is highest in **Thailand**, while **Singapore** and **Malaysia** consumers prefer **HEVs**.”



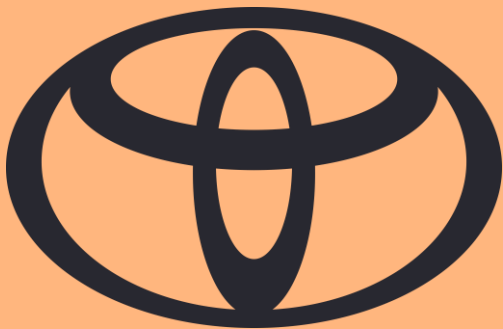
Conventional Fuel Economy Combined by Manufacturer

Top 10 - All Dimensions



Source : Electric & Alternative Fuel Vehicles [2023]

*Toyota has the highest Conventional Fuel Economy.



Source : Electric & Alternative Fuel Vehicles [2023]



Coal



Oil



Natural Gas



Kerosene

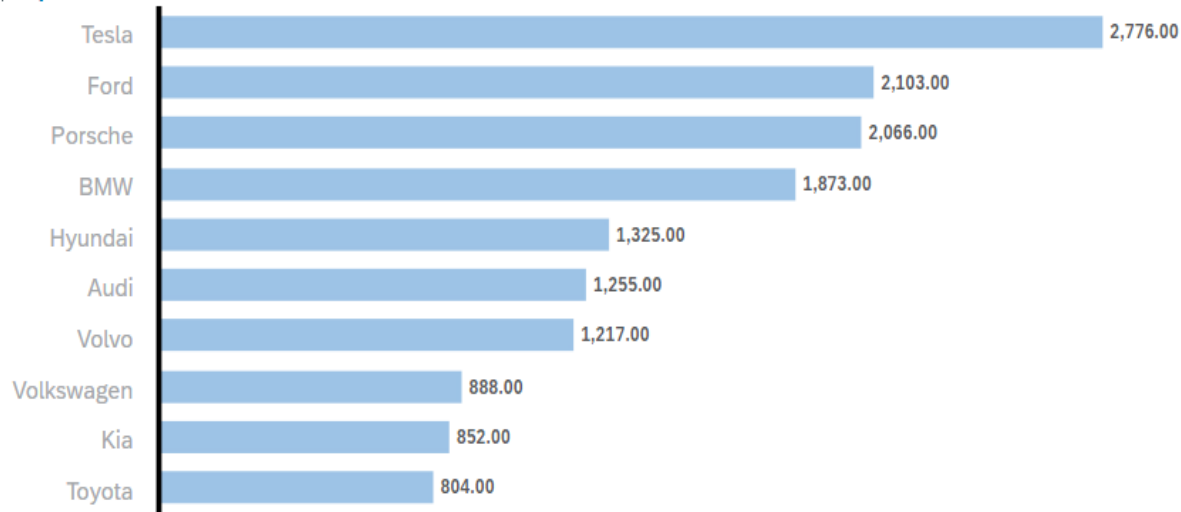


Propane

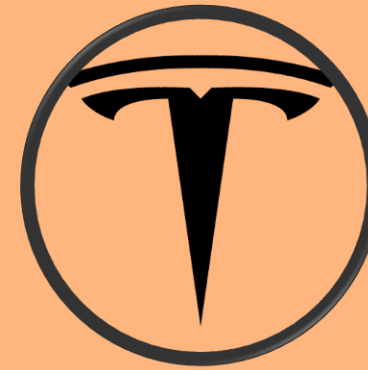


Alternative Fuel Economy Combined by Manufacturer

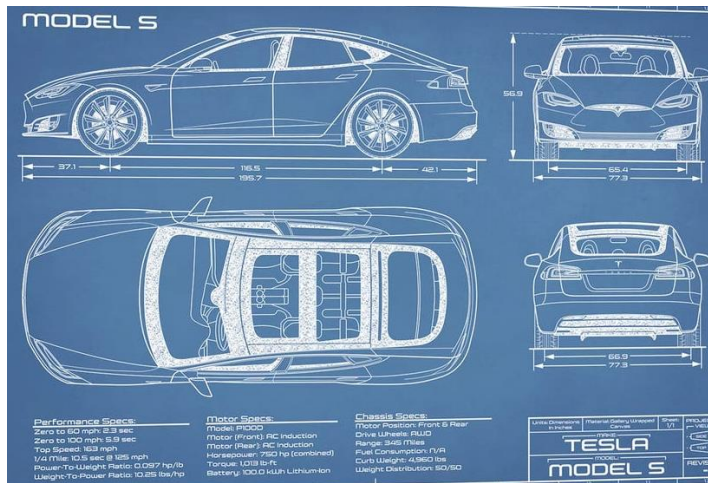
🔌 | Top 10 - All Dimensions



Source : Electric & Alternative Fuel Vehicles [2023]



Tesla is
deemed to have
the highest
**Alternative
Fuel Economy**
on the market



Source: www.tesla.com

“We design sustainable systems that are massively scalable—resulting in the greatest environmental benefit possible. Our energy generation and storage products work together with our electric vehicles to amplify their impact.”

References

ASEAN STATISTICAL YEARBOOK 2022

<https://www.aseanstats.org/publication/asyb2022/>

ASEAN Fuel Economy Roadmap for the Transport Sector 2018-2025: with Focus on Light-Duty Vehicles

<https://asean.org/wp-content/uploads/2021/08/ASEAN-Fuel-Economy-Roadmap-FINAL-2.pdf>

Greenhouse Gas Emissions from a Typical Passenger Vehicle

<https://www.epa.gov/greenvehicles/greenhouse-gas-emissions-typical-passenger-vehicle#:~:text=typical%20passenger%20vehicle%3F-,A%20typical%20passenger%20vehicle%20emits%20about%204.6%20metric%20tons%20of,8%2C887%20grams%20of%20CO2.>

EV for Anti-Pollution Matter

<https://www.gnlm.com.mm/ev-for-anti-pollution-matter/>

2022 Global Automotive Consumer Study Southeast Asia Perspectives

<https://www2.deloitte.com/vn/en/pages/consumer-business/articles/automotive-consumer-study-sea.html>

About Tesla

<https://www.tesla.com/about>